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Finite Elastic Deformations in Liquid-Saturated and Empty Porous Solids

(Dedicated to Franz Ziegler on the occasion of his 60th birthday)

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Abstract. Based on the Theory of Porous Media (TPM), a formulation of a fluid-saturated porous solid is presented where both constituents, the solid and the fluid, are assumed to be materially incompressible. Therefore, the so-called point of compaction exists. This deformation state is reached when all pores are closed and any further volume compression is impossible due to the incompressibility constraint of the solid skeleton material. To describe this effect, a new finite elasticity law is developed on the basis of a hyperelastic strain energy function, thus governing the constraint of material incompressibility for the solid material. Furthermore, a power function to describe deformation dependent permeability effects is introduced.

After the spatial discretization of the governing field equations within the finite element method, a differential algebraic system in time arises due to the incompressibility constraint of both constituents. For the efficient numerical treatment of the strongly coupled nonlinear solid-fluid problem, a consistent linearization of the weak forms of the governing equations with respect to the unknowns must be used.

Key words: deformation dependent permeability, point of compaction, finite hyperelasticity, elasticity tensor, differential algebraic equations (DAE), consistent linearization.

1. Introduction

Fluid-saturated porous solids, for example polymer foams, metal foams, sponges or biological tissues can undergo large elastic deformations in the macroscopic range. These foams consist of a porous solid skeleton which is filled with a fluid. Based on the Theory of Porous Media (TPM), these types of materials can be described by a two-phase model. The Theory of Porous Media is defined by the classical theory of mixtures combined with the concept of volume fractions. The reader who is interested in a broad review of fundamentals of this theory is referred to Bowen (1976, 1980), de Boer and Ehlers (1986) or Ehlers (1989, 1993).

In this work, a fluid-saturated solid with materially incompressible constituents is described. Incompressibility of a constituent (material incompressibility) implies incompressibility in the material's micro range, which does not lead to macroscopic incompressibility. Even if the realistic density of a constituent is constant, the partial density can still change through changes in the volume fraction. Therefore, the

so-called point of compaction exists. This deformation state is reached, when all pores are closed and further volume deformations are impossible due to the incompressibility constraint of the solid skeleton material. To describe this effect, a new finite elasticity law is developed in the framework of the hyperelasticity concept. Fluid viscosity is taken into account by the drag force, where the permeability is deformation dependent and assumed to be a function of the actual porosity of the solid skeleton.

Concerning the numerical treatment of the volumetrically coupled solid–fluid problem, incompressibility of both constituents leads to a system of differential-algebraic equations (DAE) in the time domain. This DAE system must be handled by an appropriate time integration scheme. The spatial discretization is carried out by the finite element method (FEM). Finally, a consistent linearization of the nonlinear equations guarantees quadratic convergence of the solution within the Newton procedure.

2. Preliminaries

2.1. KINEMATICS

In the TPM, a fluid-saturated porous medium can be treated as an immiscible mixture of constituents φ^α ($\alpha = \mathrm{S}$: solid phase; $\alpha = \mathrm{F}$: fluid phase). A macroscopic description implies a model of superimposed continua. Thus, at any time t each spatial point of the current configuration is simultaneously occupied by particles of both constituents, solid and fluid. These particles proceed from different reference positions $\mathbf{X}_\alpha$ at the time t_0. Thus, each constituent has its own motion function

$$\mathbf{x} = \chi_\alpha(\mathbf{X}_\alpha, t) \tag{1}$$

and its own velocity field

$$\overset{\prime}{\mathbf{x}}_\alpha = \frac{\partial \chi_\alpha}{\partial t}. \tag{2}$$

With the introduction of the volume fractions

$$n^\alpha = n^\alpha(\mathbf{x}, t), \tag{3}$$

which are defined as the local ratios between the current constituent volumes $\mathrm{d}v^\alpha$ and a current bulk volume element $\mathrm{d}v$, the saturation condition reads

$$n^\mathrm{S} + n^\mathrm{F} = 1. \tag{4}$$

Associated with each constituent φ^α is a realistic density $\rho^{\alpha\mathrm{R}}$, defined as the mass of φ^α per unit of $\mathrm{d}v^\alpha$, and a partial density ρ^α, defined as the mass of φ^α per unit of $\mathrm{d}v$. The density functions are related by the volume fractions

$$\rho^\alpha = n^\alpha \, \rho^{\alpha\mathrm{R}}. \tag{5}$$

Thus, it is obvious that incompressibility of a constituent ($\rho^{\alpha R} = \text{const.}$) does not lead to macroscopic incompressibility, because the partial density ρ^α can still change through changes in the volume fractions n^α.

For any given differentiable field function $\Gamma = \Gamma(\mathbf{x}, t)$, the material time derivatives of Γ corresponding to the individual motion functions of the respective constituent φ^α are defined by

$$\frac{\mathrm{d}_\alpha}{\mathrm{d}t}\,\Gamma = (\Gamma)'_\alpha = \frac{\partial \Gamma}{\partial t} + \operatorname{grad}\Gamma \cdot \overset{\prime}{\mathbf{x}}_\alpha, \tag{6}$$

where 'grad' denotes the partial differentiation with respect to $\mathbf{x}$.

The solid motion can be described in a Lagrangian setting by introducing the displacement vector of the solid phase

$$\mathbf{u}_S = \mathbf{x} - \mathbf{X}_S. \tag{7}$$

Thus, the solid deformation gradient is defined by

$$\mathbf{F}_S = \frac{\partial \mathbf{x}}{\partial \mathbf{X}_S} =: \operatorname{Grad}_S \mathbf{x}. \tag{8}$$

The Jacobian determinant of the deformation gradient

$$J_S = \det \mathbf{F}_S \tag{9}$$

describes the volume deformation of a volume element of the mixture in the actual configuration by $\mathrm{d}v = J_S\, \mathrm{d}V_{0S}$, where $\mathrm{d}V_{0S}$ is the volume element of the mixture in the reference configuration of the solid.

The fluid motion is described in an Eulerian setting with the seepage velocity, relating the fluid motion to the deforming solid skeleton matrix:

$$\mathbf{w}_F = \overset{\prime}{\mathbf{x}}_F - \overset{\prime}{\mathbf{x}}_S\,. \tag{10}$$

2.2. BALANCE RELATIONS

Excluding mass exchanges between the solid skeleton and the pore-fluid and neglecting acceleration terms, the mechanical balance equations for each constituent φ^α read, see, e.g., Ehlers (1993):

Balance of mass:

$$(\rho^\alpha)'_\alpha + \rho^\alpha \operatorname{div} \overset{\prime}{\mathbf{x}}_\alpha = 0. \tag{11}$$

Balance of linear momentum (quasi-static):

$$\operatorname{div} \mathbf{T}^\alpha + \rho^\alpha \mathbf{b} + \hat{\mathbf{p}}^\alpha = \mathbf{0}. \tag{12}$$

In the above relations, 'div' is the divergence operator corresponding to 'grad', $\mathbf{T}^\alpha$ is the partial Cauchy stress tensor, $\mathbf{b}$ is the body force density acting upon both constituents, and $\hat{\mathbf{p}}^\alpha$ represents the momentum production term. For two-phase materials, this term can be interpreted as the interaction force per unit of bulk volume. Thus, the constraint

$$\hat{\mathbf{p}}^{\mathrm{S}} + \hat{\mathbf{p}}^{\mathrm{F}} = \mathbf{0} \tag{13}$$

must hold, due to the fact that the sum of the constituent balance equations yields the same form as the balance equation of a single-phase material. Assuming microscopically nonpolar constituents, the partial Cauchy stress tensor $\mathbf{T}^\alpha$ is symmetric.

With the use of the incompressibility condition ($\rho^{\alpha\mathrm{R}} = \text{const.}$) and the definition of the partial densities (5), the mass balance equations lead to volume balance equations

$$(n^\alpha)'_\alpha + n^\alpha \operatorname{div} \overset{\prime}{\mathbf{x}}_\alpha = 0. \tag{14}$$

By integration of the solid volume balance relation and by use of the Jacobian determinant (9) of the deformation gradient, the solid volume fraction reads

$$n^{\mathrm{S}} = \det \mathbf{F}_{\mathrm{S}}^{-1}\, n^{\mathrm{S}}_{0\mathrm{S}} = \frac{1}{J_{\mathrm{S}}}\, n^{\mathrm{S}}_{0\mathrm{S}}, \tag{15}$$

where $n^{\mathrm{S}}_{0\mathrm{S}}$ is the solid volume fraction referring to the solid reference configuration. The incompressibility constraint for the whole binary medium (the mixture) is found by combining the volume balance equations (14) for the solid and the fluid phases together with the saturation constraint (4):

$$\operatorname{div}\left(n^{\mathrm{F}}\, \mathbf{w}_{\mathrm{F}} + (\mathbf{u}_{\mathrm{S}})'_{\mathrm{S}}\right) = 0. \tag{16}$$

2.3. BASIC CONSTITUTIVE EQUATIONS

The binary model with incompressible constituents has to be completed by constitutive assumptions for the partial solid and fluid stress tensors, $\mathbf{T}^{\mathrm{S}}$ and $\mathbf{T}^{\mathrm{F}}$, as well as for the interaction term (the drag force) $\hat{\mathbf{p}}^{\mathrm{F}} = -\hat{\mathbf{p}}^{\mathrm{S}}$. In the present section, we refer to results given by Ehlers (1993). Due to the incompressibility constraints of the constituents, each of the constitutive equations consists of two terms. The first term includes the effective fluid pressure p, which could be interpreted as a Lagrangian multiplier to govern the incompressibility requirement. The second term represents the so-called extra quantity, characterized by the additional index $(..)_{\mathrm{E}}$. Therefore, the stress tensors are given by

$$\mathbf{T}^\alpha = -n^\alpha p\, \mathbf{I} + \mathbf{T}^\alpha_{\mathrm{E}}, \tag{17}$$

where $\mathbf{I}$ is the second order identity tensor. The interaction force is given by

$$\hat{\mathbf{p}}^{\mathrm{F}} = p \operatorname{grad} n^{\mathrm{F}} + \hat{\mathbf{p}}^{\mathrm{F}}_{\mathrm{E}}. \tag{18}$$

As is usual in hydraulics, it is assumed that the friction stress is negligible:

$$\mathbf{T}_{\mathrm{E}}^{\mathrm{F}} \approx \mathbf{0}. \tag{19}$$

The internal friction between the solid and the fluid constituents is then taken into account by the extra interaction force, which, in the case of isotropic permeability, is given by

$$\hat{\mathbf{p}}_{\mathrm{E}}^{\mathrm{F}} = -\frac{(n^{\mathrm{F}})^2\,\gamma^{\mathrm{FR}}}{k^{\mathrm{F}}}\ \mathbf{w}_{\mathrm{F}}. \tag{20}$$

Therein, γ^{FR} is the effective weight of the fluid and k^{F} is the Darcy permeability parameter, which is a macroscopic quantity that depends both on the viscosity of the fluid and on the size and structure of the pore space. From the balance of momentum (12) for the fluid phase in combination with the constitutive assumptions for the fluid, (19) and (20), the relation

$$n^{\mathrm{F}}\,\mathbf{w}_{\mathrm{F}} = -\frac{k^{\mathrm{F}}}{\gamma^{\mathrm{FR}}}\ \left(\operatorname{grad} p - \rho^{\mathrm{FR}}\,\mathbf{b}\right) \tag{21}$$

holds, which is known as Darcy's law.

The constitutive assumptions for the solid extra stress tensor $\mathbf{T}_{\mathrm{E}}^{\mathrm{S}}$ will be discussed in greater detail in Section 3.

2.4. DEFORMATION DEPENDENT DARCY PERMEABILITY PARAMETER

In general, the Darcy permeability parameter k^{F}, which relates the filter velocity $(n^{\mathrm{F}}\,\mathbf{w}_{\mathrm{F}})$ to the pressure gradient grad p, is deformation dependent. The value of k^{F} decreases when the pore volume decreases and must be zero when all pores are closed. In this case, no fluid flow is possible. It is assumed that the permeability depends on the volume fraction of the fluid, which represents the porosity of the solid skeleton in the saturated two-phase model. The fluid volume fraction is given by the saturation constraint (4) and with the aid of the Jacobian determinant:

$$n^{\mathrm{F}} = 1 - n^{\mathrm{S}} = 1 - \frac{1}{J_{\mathrm{S}}}\,n_{0\mathrm{S}}^{\mathrm{S}}. \tag{22}$$

The deformation dependency of the Darcy parameter is described by a power function

$$k^{\mathrm{F}}(n^{\mathrm{F}}) = k_0^{\mathrm{F}}\left(\frac{n^{\mathrm{F}}}{n_{0\mathrm{S}}^{\mathrm{F}}}\right)^{\kappa}, \tag{23}$$

where k_0^{F} is the initial permeability at the initial porosity $n_{0\mathrm{S}}^{\mathrm{F}} = 1 - n_{0\mathrm{S}}^{\mathrm{S}}$. The parameter κ governs the strength of the deformation dependency. For $\kappa = 0$, k^{F} is constant.

3. A New Finite Elasticity Law

We proceed from a hyperelastic strain energy function $W^{\mathrm{S}} = \rho_{0\mathrm{S}}^{\mathrm{S}}\psi^{\mathrm{S}}$, where ψ^{S} is the partial free energy density of the solid. For isotropic materials, the strain energy

function must be a function of the principal invariants or eigenvalues of the right or left Cauchy–Green deformation tensors

$$\mathbf{C}_S = \mathbf{F}_S^T \, \mathbf{F}_S, \qquad \mathbf{B}_S = \mathbf{F}_S \, \mathbf{F}_S^T, \tag{24}$$

see for example Ogden (1984). The elasticity law describes the deformation of the porous solid skeleton structure. Thus, the use of a compressible elasticity law is necessary, even if material incompressibility of the solid skeleton is assumed, since large volumetric strains can occur as a result of porosity changes (changes of the volume fractions n^S and n^F). Therefore, the material parameters included in the following elasticity law, in this work the Lamé constants μ^S and Λ^S, are macroscopical parameters of the solid skeleton structure and not of the microscopic skeleton material itself.

In what follows, a strain energy function of the Neo-Hookean type (Rivlin, 1948) with an extension term for the description of compressibility effects is used:

$$W^S = \tfrac{1}{2}\,\mu^S\,(I_1 - 3) - \mu^S \ln J_S + U^S(J_S). \tag{25}$$

Therein, I_1 is the first principal invariant of $\mathbf{C}_S$ or $\mathbf{B}_S$. The second term in (25) guarantees that the reference configuration is stress free. The extension function $U^S(J_S)$ describes the finite volume changing domain including the concept of the compaction point.

Recall that in the case of an incompressible matrix material, the point of compaction is reached when all pores are closed and further volume deformations are impossible due to the incompressibility constraint of the solid material. On the other hand, the pores can be widened infinitely, allowing large volume extensions. Thus, with (15) and $0 \leqslant n^S \leqslant 1$, the Jacobian determinant of $\mathbf{F}_S$ is limited by

$$n_{0S}^S < J_S < \infty. \tag{26}$$

This effect must be expressible by the elasticity law. Therefore, the constitutive formulation of U^S is restricted to certain physical requirements. Under hydrostatic loading, the point of compaction can only be reached with an infinite strain energy U^S and an infinite pressure resulting from the hydrostatic Cauchy stress $\partial U^S / \partial J_S$. Thus, the conditions

$$\left.\begin{aligned} U^S &\to +\infty, \\ \frac{\partial U^S}{\partial J_S} &\to -\infty, \end{aligned}\right\} \quad \text{for} \quad J_S \to n_{0S}^S \tag{27}$$

must hold. Furthermore, the usual definition of the Lamé parameter

$$\Lambda^S = \left.\frac{\partial^2 U^S}{\partial J_S^2}\right|_{J_S=1} \tag{28}$$

must be satisfied. This guarantees a transition to an elasticity law of the Hookean type in the case of a linearization procedure. If the initial volume fraction n_{0S}^S declines, the

reference solid volume decreases, with an increasing pore space. Thus, if $n^{\mathrm{S}}_{0\mathrm{S}} \to 0$, the deformation state associated with the point of compaction reduces to a singular point. In this case, the elasticity law should formally coincide with an elasticity law of a compressible nonporous solid, which allows for this deformation state. Furthermore, for $n^{\mathrm{S}}_{0\mathrm{S}} \to 1$ a 'one-sided' incompressible material law must result. In this case, a volume compression is not possible, whereas a volume expansion by increasing of the pores can occur. A further requirement is that U^{S} is a convex function of J_{S},

$$\frac{\partial^2 U^{\mathrm{S}}}{\partial J_{\mathrm{S}}^2} > 0, \tag{29}$$

thus guaranteeing uniqueness of the material model.

Based on the function U^{S} found by Ogden (1972) in the context of finite elasticity for non-porous solids, a modification of this term is presented meeting the above requirements. For the description of a porous solid skeleton consisting of an incompressible matrix material, the modified function can be derived by

$$U^{\mathrm{S}}(J_{\mathrm{S}}) = \Lambda^{\mathrm{S}}\,(1 - n^{\mathrm{S}}_{0\mathrm{S}})^2 \left(\frac{J_{\mathrm{S}} - 1}{1 - n^{\mathrm{S}}_{0\mathrm{S}}} - \ln \frac{J_{\mathrm{S}} - n^{\mathrm{S}}_{0\mathrm{S}}}{1 - n^{\mathrm{S}}_{0\mathrm{S}}} \right), \tag{30}$$

see Ehlers and Eipper (1997). However, the function U^{S} from (30) is not only valid in combination with the Neo-Hookean model (25). It is possible as well to combine the function U^{S} with other known strain energy functions, for example, the Ogden model (Ogden, 1972) formulated in eigenvalues of the deformation tensors.

In the case of hyperelasticity, the Kirchhoff extra stress tensor, which is a weighted Cauchy stress tensor ($\boldsymbol{\tau}^{\mathrm{S}}_{\mathrm{E}} = J_{\mathrm{S}}\,\mathbf{T}^{\mathrm{S}}_{\mathrm{E}}$), can be obtained by

$$\boldsymbol{\tau}^{\mathrm{S}}_{\mathrm{E}} = \mathbf{F}_{\mathrm{S}} \left(2\,\frac{\partial W^{\mathrm{S}}}{\partial \mathbf{C}_{\mathrm{S}}} \right) \mathbf{F}^{\mathrm{T}}_{\mathrm{S}} \tag{31}$$

$$= 2\,\frac{\partial W^{\mathrm{S}}}{\partial \mathbf{B}_{\mathrm{S}}}\,\mathbf{B}_{\mathrm{S}}. \tag{32}$$

For the Neo-Hookean model (25) in combination with U^{S} from Equation (30), the extra stress tensor reads

$$\boldsymbol{\tau}^{\mathrm{S}}_{\mathrm{E}} = \mu^{\mathrm{S}}\,(\mathbf{B}_{\mathrm{S}} - \mathbf{I}) + \Lambda^{\mathrm{S}}(1 - n^{\mathrm{S}}_{0\mathrm{S}})^2 \left(\frac{J_{\mathrm{S}}}{1 - n^{\mathrm{S}}_{0\mathrm{S}}} - \frac{J_{\mathrm{S}}}{J_{\mathrm{S}} - n^{\mathrm{S}}_{0\mathrm{S}}} \right) \mathbf{I}. \tag{33}$$

The Oldroyd stress rate (lower Lie derivative of $\boldsymbol{\tau}^{\mathrm{S}}_{\mathrm{E}}$) is defined by

$$(\boldsymbol{\tau}^{\mathrm{S}}_{\mathrm{E}})^{\triangledown} = (\boldsymbol{\tau}^{\mathrm{S}}_{\mathrm{E}})'_{\mathrm{S}} - \mathbf{L}_{\mathrm{S}}\,\boldsymbol{\tau}^{\mathrm{S}}_{\mathrm{E}} - \boldsymbol{\tau}^{\mathrm{S}}_{\mathrm{E}}\,\mathbf{L}^{\mathrm{T}}_{\mathrm{S}}, \tag{34}$$

where $\mathbf{L}_{\mathrm{S}} = \operatorname{grad} \overset{\prime}{\mathbf{x}}_{\mathrm{S}}$ is the spatial velocity gradient. The elasticity tensor $\overset{4}{\mathbf{C}}$ of the actual configuration relates the Oldroyd stress rate to the solid deformation rate $\mathbf{D}_{\mathrm{S}} = 1/2(\mathbf{L}_{\mathrm{S}} + \mathbf{L}^{\mathrm{T}}_{\mathrm{S}})$:

$$(\boldsymbol{\tau}^{\mathrm{S}}_{\mathrm{E}})^{\triangledown} =: \overset{4}{\mathbf{C}}\,\mathbf{D}_{\mathrm{S}}. \tag{35}$$

Therein, $\overset{4}{\mathbf{C}}$ can be derived from the elastic tangent of the reference configuration

$$\overset{4}{\mathbf{B}} = 4\,\frac{\partial^2 W^{\mathrm{S}}}{\partial \mathbf{C}_{\mathrm{S}} \otimes \partial \mathbf{C}_{\mathrm{S}}} \tag{36}$$

by the fourth-order push-forward transport mechanism

$$\overset{4}{\mathbf{C}} = (\mathbf{F}_{\mathrm{S}} \otimes \mathbf{F}_{\mathrm{S}})^{\overset{23}{\mathrm{T}}} \;\overset{4}{\mathbf{B}}\; (\mathbf{F}_{\mathrm{S}}^{\mathrm{T}} \otimes \mathbf{F}_{\mathrm{S}}^{\mathrm{T}})^{\overset{23}{\mathrm{T}}}. \tag{37}$$

Therein, the transposition $(..)^{\overset{ik}{\mathrm{T}}}$ indicates an exchange of the ith and kth basis system of higher order tensors. For the model under consideration, the elasticity tensor is given by the following isotropic representation:

$$\overset{4}{\mathbf{C}} = 2\alpha(\mathbf{I} \otimes \mathbf{I})^{\overset{23}{\mathrm{T}}} + \beta(\mathbf{I} \otimes \mathbf{I}), \tag{38}$$

where

$$\begin{aligned}
\alpha(J_{\mathrm{S}}) &= \left(\mu^{\mathrm{S}} - J_{\mathrm{S}}\frac{\partial U^{\mathrm{S}}}{\partial J_{\mathrm{S}}}\right) \\
&= \left[\mu^{\mathrm{S}} - \Lambda^{\mathrm{S}}(1 - n_{0S}^{\mathrm{S}})J_{\mathrm{S}}\;\frac{J_{\mathrm{S}} - 1}{J_{\mathrm{S}} - n_{0S}^{\mathrm{S}}}\right], \\
\beta(J_{\mathrm{S}}) &= \left(J_{\mathrm{S}}\,\frac{\partial U^{\mathrm{S}}}{\partial J_{\mathrm{S}}} + J_{\mathrm{S}}^2\,\frac{\partial^2 U^{\mathrm{S}}}{\partial J_{\mathrm{S}}^2}\right) \\
&= \Lambda^{\mathrm{S}}\left[(1 - n_{0\mathrm{S}}^{\mathrm{S}})\;J_{\mathrm{S}}\;\frac{J_{\mathrm{S}}^2 - 2\,n_{0\mathrm{S}}^{\mathrm{S}}\,J_{\mathrm{S}} + n_{0\mathrm{S}}^{\mathrm{S}}}{(J_{\mathrm{S}} - n_{0\mathrm{S}}^{\mathrm{S}})^2}\right];
\end{aligned} \tag{39}$$

$(\mathbf{I} \otimes \mathbf{I})^{\overset{23}{\mathrm{T}}}$ represents the fourth order identity tensor.

The described elasticity law is also suitable for the computation of empty porous solid skeleton matrices, that is, porous materials, where the mechanical description of the fluid constituent can be neglected.

4. Weak Form and FEM-Equations

For the numerical treatment of the problem within the finite element method (FEM), the weak forms of the field equations are used. Therefore, the governing equations are multiplied by independent test functions and are then integrated over the spatial domain $\mathcal{B}$ with boundary $\partial\mathcal{B}$.

The balance equation of momentum (12) for the binary medium under consideration together with the constitutive Equations (17) and (19), are transformed into the weak formulation

$$\int_{\mathcal{B}} (\mathbf{T}_{\mathrm{E}}^{\mathrm{S}} - p\,\mathbf{I}) \cdot \operatorname{grad} \delta\mathbf{u}_{\mathrm{S}}\;\mathrm{d}v - \int_{\mathcal{B}} (\rho^{\mathrm{S}} + \rho^{\mathrm{F}})\,\mathbf{b} \cdot \delta\mathbf{u}_{\mathrm{S}}\;\mathrm{d}v = \int_{\partial\mathcal{B}} \mathbf{t} \cdot \delta\mathbf{u}_{\mathrm{S}}\;\mathrm{d}a, \tag{40}$$

where $\delta\mathbf{u}_S$ is the test function corresponding to the solid displacement and $\mathbf{t}$ is the surface tension. The solid extra stress tensor $\mathbf{T}^S_E$ is computed from the elasticity law presented in the previous section.

Considering quasi-static problems, the seepage velocity $\mathbf{w}_F$ can be eliminated from the field equations. Inserting Darcy's law (21) into the volume balance of the mixture (16), the weak form of the volume balance reads

$$\int_{\mathcal{B}} \operatorname{div}(\mathbf{u}_S)'_S \, \delta p \, \mathrm{d}v + \int_{\mathcal{B}} \frac{k^F}{\gamma^{FR}} \left(\operatorname{grad} p - \rho^{FR}\mathbf{b}\right) \cdot \operatorname{grad} \delta p \, \mathrm{d}v = -\int_{\partial\mathcal{B}} q \, \delta p \, \mathrm{d}a, \tag{41}$$

where δp is the test function corresponding to p, $(\mathbf{u}_S)'_S$ is the displacement velocity of the solid and $q = n^F \mathbf{w}_F \cdot \mathbf{n}$ is the filter volume flow of the fluid draining through the surface $\partial\mathcal{B}$.

In a standard Galerkin procedure (see e.g. Zienkiewicz and Taylor, 1984), the unknown fields $\mathbf{u}_S$ and p and the corresponding test functions $\delta\mathbf{u}_S$ and δp are approximated by the same shape functions. In the present investigation, quadratic shape functions for the approximation of the solid displacement and linear shape functions for the approximation of the pore pressure are used.

Applying a spatial discretization, the global system of equations reads

$$\mathbf{G} = \mathbf{M}\frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t} + \mathbf{k}(\mathbf{y}) - \mathbf{f}(t) = \mathbf{0}. \tag{42}$$

Therein, the vector $\mathbf{y}$ of the unknowns contains the values of the solid displacement and the pore pressure at the FEM-nodes. $\mathbf{M}$ is the generalized mass matrix, $\mathbf{k}$ the generalized stiffness vector, and $\mathbf{f}$ is the vector of the generalized forces. Due to the incompressibility constraint, the pore pressure has no evolution equation and the matrix $\mathbf{M}$ is singular. Therefore, (42) is an index one system of DAE of first order in time. For the time integration of the DAE, an implicit Euler integration scheme can be used, see Brenan *et al.* (1989).

Time discretization of Equation (42) leads to a system of nonlinear equations. For the solution, a Newton procedure is applied in each time step. For an efficient treatment, it is necessary to apply a consistent linearization of the weak forms with respect to the unknowns, see Marsden and Hughes (1983), Wriggers (1988). This guarantees quadratic convergence within a Newton procedure. A consistent linearization of the weak forms (40) and (41) with respect to the unknown field functions $\mathbf{u}_S$ and p leads, after a spatial discretization, to a global tangent operator $\mathrm{D}\mathbf{G}$ of (42), where the elasticity tensor (38) plays an important role. Thus, in each time step n, a Newton iteration (iteration step k)

$$\mathrm{D}\mathbf{G}^k_{n+1} \, \Delta\mathbf{y}^k_{n+1} = -\mathbf{G}^k_{n+1} \tag{43}$$

must be computed, until convergence is reached. The solution vector is updated by the incremental solution vector $\Delta\mathbf{y}$:

$$\mathbf{y}^{k+1}_{n+1} = \mathbf{y}^k_{n+1} + \Delta\mathbf{y}^k_{n+1}. \tag{44}$$

5. Examples

5.1. COMPRESSION OF A POROUS TUBE: POINT OF COMPACTION

The first example handles a fluid-saturated porous tube, where the upper boundary is perfectly drained and loaded by a time dependent external force. The other boundaries are rigid and undrained (see Figure 1). In this one-dimensional deformation problem, the point of compaction can be calculated analytically. The height of the tube is 1 m and the volume fraction of the solid in the reference configuration is $n^{\mathrm{S}}_{0\mathrm{S}} = 0.67$. Therefore, the point of compaction is reached when the subsidence of the surface has the amount of 0.33 m. The other material parameters were chosen as follows: $\mu^{\mathrm{S}} = 5583$ kN/m^2, $\Lambda^{\mathrm{S}} = 8375$ kN/m^2 and $\gamma^{\mathrm{FR}} = 10^4$ N/m^3. In this example, the Darcy permeability parameter is assumed to be deformation independent and is chosen to $k^{\mathrm{F}}_0 = 10^{-4}$ m/s. The body force density **b** is neglected. Figure 1 shows the typical time-deformation history of a consolidation process, where a step loading of $f = 7500$ kN/m^2 is applied (solid line). The viscosity of the fluid leads to a time dependent deformation under the step loading. While the consolidation process is running, fluid is drained out through the upper surface. The maximum deformation is reached, when the consolidation process is finished.

For the validation of the finite hyperelastic law presented in Section 3, a second load case is applied. After the same step loading at time $t_0 = 0$, as in the first load case, the external load is increased in time until it has reached 20 times the initial step loading. The deformation of the surface asymptotically approaches the point of compaction (dashed line). In this deformation state, the total pore-fluid is drained

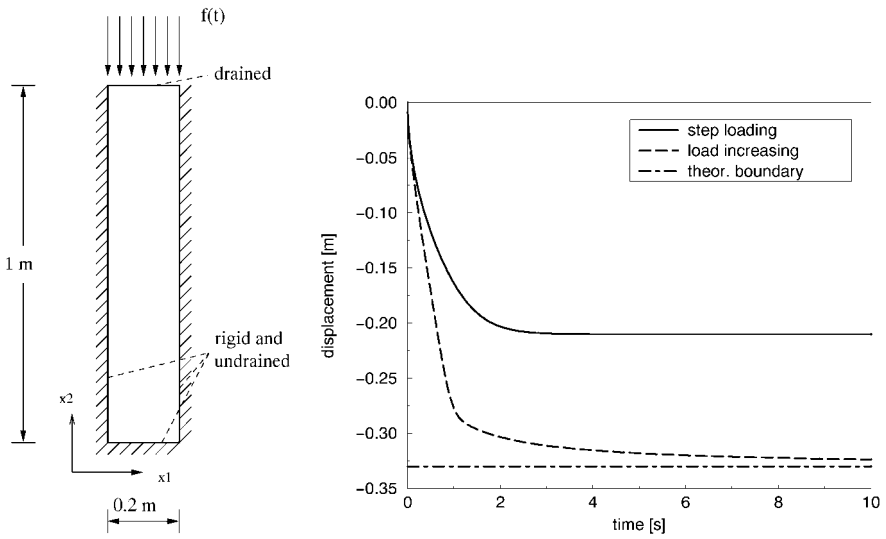


Figure 1. Boundary conditions and displacement of the surface under step loading and under load increasing.

out, and the remaining solid body consists of an incompressible material. Thus, no further volume deformation is possible, and the tube locks.

This result cannot be obtained with finite elasticity laws of usual nonporous solid materials. In this case, the computation of the porous tube would lead to physically impossible results. The point of compaction would be surpassed and the saturation constraint would only be fulfilled with negative volume fractions of the fluid, which is obviously physically impossible.

5.2. COMPRESSION OF A POROUS TUBE: DEFORMATION DEPENDENT PERMEABILITY

Now, the influence of the deformation dependent Darcy permeability to the time scale of the time-deformation history in the consolidation problem should be examined. Therefore, the same boundary value problem as in the first example, the compression of a porous cube under a step loading, is used. The material parameters are also the same as in the first example, except that the permeability k^{F} is not constant. Here, the power function (23) for the Darcy parameter is used, where the exponent κ is varied.

Figure 2 shows that the time scale in the consolidation problem changes significantly, when the parameter κ is changed. The higher the exponent κ is, the more the permeability in the volume compression range decreases, and therefore it takes longer until the consolidation process is finished.

5.3. CONSOLIDATION OF A CUBE

In this example, a quarter of the upper surface of a fluid-saturated porous cube is left undrained and loaded by a ramp load. The remainder of the upper surface is unloaded and perfectly drained. The spatial discretization of the problem is based on

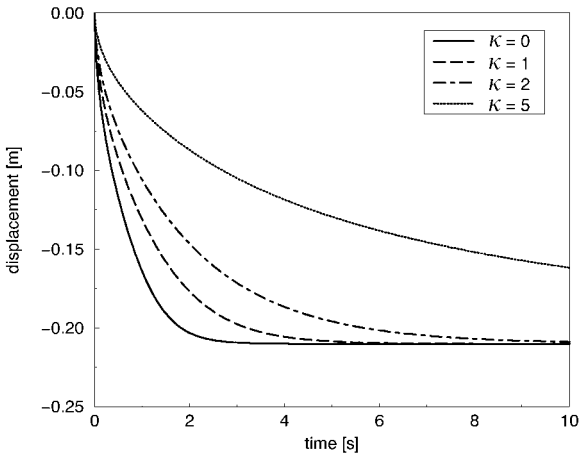


Figure 2. Displacement of the tube surface under step loading, with different parameters κ in the deformation dependent Darcy permeability parameter.

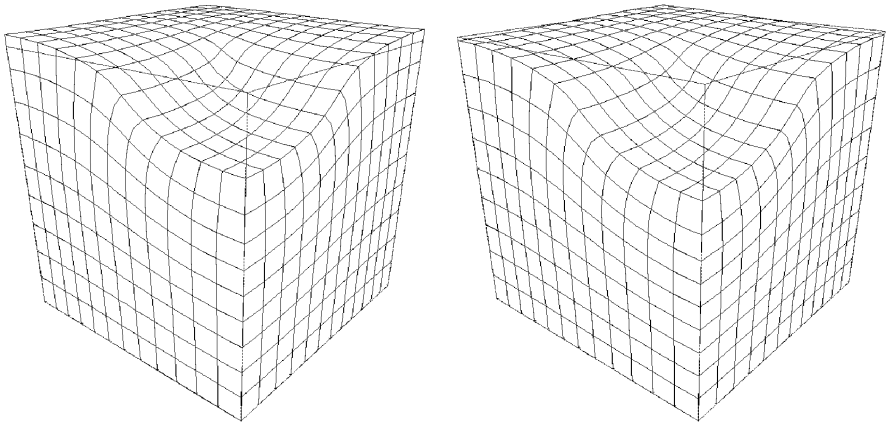


Figure 3. Deformation of the cube after applying external load (left) and when consolidation process is finished (right).

20-node 3-d brick elements. Due to the viscosity of the fluid, the deformation process is time dependent. Directly after having applied the full external load, the unloaded corners of the cube vault up. While the consolidation process is running, the fluid is drained out through the (drained) upper surface of the cube. Therefore, a subsidence of the unloaded corners under the original form of the cube is only reached when the consolidation process is finished, see Figure 3.

6. Conclusion

Proceeding from the TPM, the model equations for a fluid-saturated solid skeleton have been derived. Even if material incompressibility of the solid phase is assumed, large volumetric strains may occur as a result of porosity changes. Therefore, the point of compaction cannot be passed. This deformation state is reached when all pores are closed and further volume compression is impossible. To describe this effect, an extension of a hyperelastic strain energy function has been developed, in which all important physical requirements are satisfied. Combining this extension term with the well known Neo-Hookean model, the extra stress tensor and the consistent elastic tangent have been derived for possible finite deformations.

In the case of finite deformations, the influence of the deformation dependent permeability on the solution of a boundary value problem is significant. Therefore, a power function describing the dependency of the Darcy permeability parameter from the porosity of the solid skeleton has been introduced. Further investigations must be done for the description of anisotropic permeability effects, induced by isochoric deformations.

As a result of the assumed incompressibility of the solid and fluid phases, spatial discretization of the weak forms of the governing equations leads to a system of DAE in time. After a time discretization, a system of nonlinear equations arises.

For the efficient numerical treatment of the strongly coupled solid–fluid problem, a consistent linearization of the governing equations has been applied.

In the numerical examples it has been shown that the viscosity of the fluid leads to a time dependent deformation under a step loading, which is known as the consolidation problem. The effectiveness of the developed elasticity law has been shown in numerical examples. A description of the point of compaction is impossible with known material laws for nonporous solids. Their application leads to physically impossible results.

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